

Manual

Calculation of Group Bonuses LLE-BGP 8.1

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1 LLE-BGP - Overview

Purpose

To identify group bonuses, the module *Calculation of Group Bonuses* complements the basic module *Calculation of Premium/Incentive Wages*. Different methods are used here to group persons. The system calculates and shows daily and monthly results for the groups. The incentive wage is assigned on a monthly basis to the persons. It is calculated using the attendance time of the person in the separate premium groups.

This module also provides the possibility to flexibly react to special situations. Example: You can use bonuses to react to unforeseen events in the production. You can also correct the wage calculation.

Integration

The module is based on the order- and person-related data of the order data collection. Optionally, you can also integrate the times of the Personnel Time Management. This data can also be imported and processed from external systems via interfaces.

Features

The Application Service (AS) to calculate group bonuses in the form of incentive and utilization bonuses provides the following features:

- Functions to define and edit group bonuses
- Integration of the incentive wage indicators in the HR master data, which can be saved in versions
- Integration of incentive wage indicators in the master data of machines and workplaces
- Optional recording of premium groups when orders are logged on
- Optional: After authorization check, employees can enter bonuses and deductions on the BDE terminal
- Display of an information on the group incentives on the PZE terminal (Windows-based).
- Interface to transfer the results to the payroll system in HYDRA standard format
- Extensive data security and safety functions: function authorizations and protection of data of premium groups via responsibility areas

Note on the terminal info group incentives

If you use the Time and Attendance (PZE) and the data collection software CTAIP or CTWIN, an info is displayed on the terminal where Time and Attendance is recorded. The terminal info displays the current account balances.

If you use LLE-BGP "Calculation of group bonuses", this info displays additional rows that include the times of the person in the premium groups and the performance level of the premium groups for the current and the previous month. For more information on this info function, refer to the document describing the HR functions of the data collection software (status 2018: documents AIP-HRF and AIP-HRL).

You can change or disable the info display as part of the customizing of the incentive wage module.

2 Formation of Groups at Group Incentives

Summary

This document describes how entered data can be assigned to the premium groups of the incentive wage determination.

- Groups are mostly formed via the assignment of machines to premium groups. The data recorded at the machines will then be assigned to the corresponding premium group. This static assignment of the machines to the premium groups is made at the client using the *Assignment of premium groups* application.
- Instead of the static assignment of machines to the premium groups, it is also possible to record a premium group at the *terminal when an order is entered in the entry dialog*. All the postings connected to the order logon will then be assigned to the premium group. At one point in time a machine can only be assigned to a single premium group, also if several orders are logged-in to the machine.
- Personal data (from the shop floor data collection and the personnel time management, if any) may as an option be assigned via the client application *Change of groups* to the premium group and this assignment will be personal and be made with the precise time. In this case, however, only the personnel postings (B-records) from the shop floor data collection can be compensated by the method of the time tickets to the premium groups. To do so, the additional function "formula-based premium/ incentive wage" and a corresponding customizing through MPDV will be required.

Order data via the machine used for posting



In the BDE shop floor data collection the premium groups will only be processed for such workplaces/ machines marked by the incentive wage indicator "G" for "group piecework"!

Per operation log-in the operation posting (U- or E-record) and the related personnel postings (B-records) will be assigned to a premium group. Since it is possible that it is worked on several premium groups at one workplace, both the entry of the premium group during the log-in of the operation and the assignment of premium groups to workplaces will be possible. To do so, HYDRA uses the following logic:

1. If a premium group is specified in the login dialog when an operation is logged-in, this premium group will be used.

2. If no premium group is specified when an operation is logged-in, the premium group assigned to the workplaces will be used.
3. If no premium group is specified when an operation is logged-in and if no premium group is assigned to the workplace/ the machine, an error message will be issued; an operation cannot be logged-in without indication of the premium group.

The dialog configuration can be used to control (per terminal) whether the field "premium group" will be available when an operation is logged in or not.

If several persons work on the same operation on group workplaces, the first person will define the premium group. The following persons must not indicate another premium group than the first person.

The assigned premium groups will then be stored for further processing in the order and personnel postings. The editing of postings can be used to correct incorrectly assigned premium groups.

Dialog configuration for entry purposes at the login of OPs

An entry via an OP log-in will only be made if the workplace has the premium indicator "G" and when the module "Calculation of group incentives" is licensed.

In this case the field for the entry of the premium group (LPRGRP) must be configured into the dialog to log-in operations (customizing).

To use this described function, the dialogs Log OP on (A_AN) and Log person on (A_P_AN) must be extended by the entry field "premium group".

The bar codes for premium groups will also be detected in those instances, in which the entry focus is on a different field.

Structure of the bar codes

A bar code starts and ends with an asterisk "*" as start and stop sign.

The bar codes must be completed to **nine** digits useful length (between the asterisks). Due to this amount of numbers the terminal will detect that this scanned bar code is a premium group.

The (shorter) premium group must be completed to **nine** digits: For **filling** purposes and for **blanks** within the premium groups **underscores** "_" must be used!

The bar code must then be formatted as bar code "Code 39" in the font "Codedreineun". This font is available with all console PCs under Windows. It can be used to create tables with bar codes for premium groups in all current Office applications.

The following characters are supported by the bar code "Code 39":

- The numbers 0 to 9
- Capital letters A to Z without umlauts
- Blanks (to express bar codes to be written as underscore "_", see examples).
- Special indicators \$ / - + . %

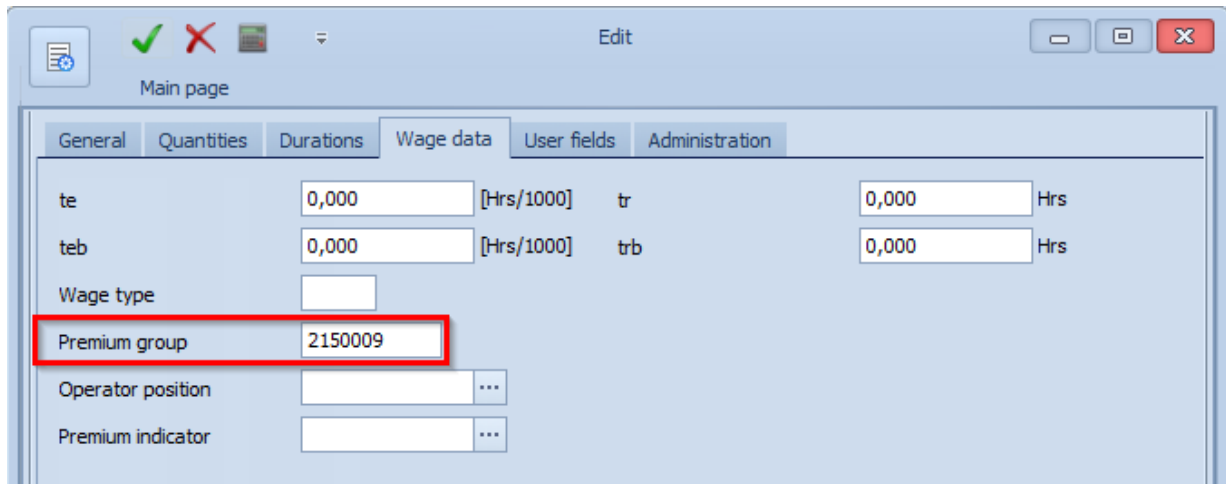
Other characters such as lower cases will **not** be supported by the bar code "Code 39".

Examples:

Premium group	Barcode content 12345678901	Barcode (font Codedreineun)
12345	*12345_____*	*12345_____*
PG 12	*PG_12_____*	*PG_12_____*
73	*73_____*	*73_____*

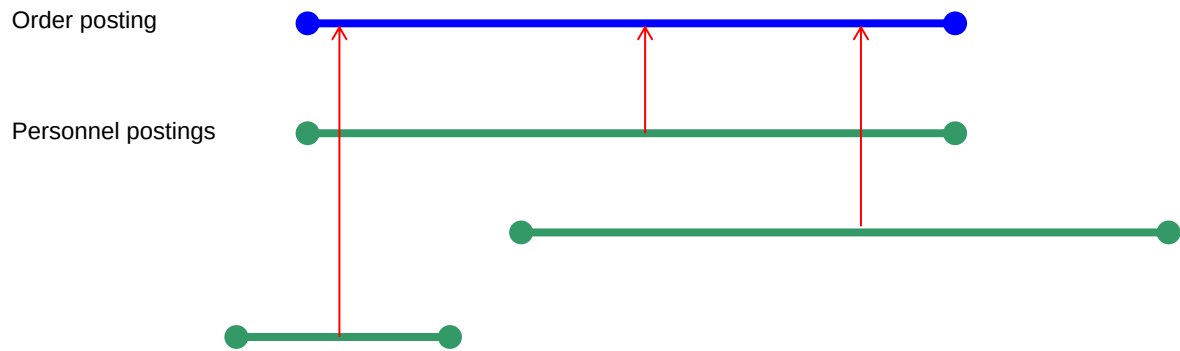
Maintenance of the premium group in the order-related postings

When BDE data are recorded, they will be assigned to premium groups depending on the system configuration. The premium group is a data field in the BDE postings and is displayed in the "wage data" tab and can also be modified there.



The screenshot shows a software window titled 'Edit' with a 'Main page' tab. The 'Wage data' sub-tab is active. It contains several input fields: 'te' (0,000 [Hrs/1000]), 'tr' (0,000 Hrs), 'teb' (0,000 [Hrs/1000]), 'trb' (0,000 Hrs), 'Wage type' (empty), 'Premium group' (2150009, highlighted with a red rectangle), 'Operator position' (empty with a dropdown arrow), and 'Premium indicator' (empty with a dropdown arrow).

When the premium group is changed in the editing of postings, HYDRA will ensure that order and personnel postings that belong together will have the same premium group. The system will recognize that order and personnel postings belong together when the time-related centers of the personnel postings are within the order posting.



3 Basics of the Group Bonus

Summary

The group-related incentive wages are used to comprise machines and/or workplaces to so called "premium groups".

By default, either a payment scheme depending on the utilization or depending on the performance efficiency rate can be assigned to a premium group. For the utilization bonus the efficiency of the machine usage time is decisive, and for the performance efficiency rate the produced quantity related to the standard time.

When the formula-based premium/ incentive wage is used, it is also possible to realize by customer-specific formulas also other calculation methods and premium modes.

Upon the login to a machine and/or workplace, the order processing data and employees will be assigned to a premium group for the processing duration.

The corresponding activities are assigned once per month to the persons on a pro-rated base to the presence times that the persons worked in the individual premium groups.

Calculation and availability of the data

The day data on premium groups will be determined and compressed through the wage calculation. These day data serve as basis for the monthly results of the premium groups. Modifications to the ADE postings will be transferred to the premium group results during the daily night run.

HYDRA will maintain these daily group results in the time domain that can be specified in the data management and automatically delete them once this period has elapsed.

The monthly results of the premium groups are calculated directly when the list of the current day data is called.

Calculation of the incentive bonus

The incentive bonus is based on the quantities that were produced in the group to one specific order. All order and personnel postings are used for the calculation of the incentive bonus that were collected at the machines of a premium group in a given settlement period (month).

The product of the quantity of interruption and end postings and of operation-related standard time (t_i and t_e) per piece is the total standard time. The ratio of this standard time to the really needed personal processing time of the group defines the group's performance.

The incentive wage determination is based on the following method:

$$\text{Performance level LG} = \frac{\sum_{ops} t_r + m * t_e + \sum Bonuses_{Targettime}}{\sum_{Persons} Processing\ times - \sum Bonuses_{Actual\ time}}$$

- The performance level is determined per premium group.
- t_r is the set-up standard time for the operation that was stored to the HYDRA order backlog at the time of processing. If necessary, this value will be entered via the PPS interface and can then be overwritten in the corresponding field of the HYDRA order backlog. The value will be used from the order postings.
- t_e is the individual standard time for the operation that was stored to the HYDRA order backlog at the time of processing. If necessary, this value will be entered via the PPS interface and can then be overwritten in the corresponding field of the HYDRA order backlog. The value will be used from the order postings.
- m is the yield that was posted during the processing of the operation in HYDRA (if necessary through several persons). The value will be used from the order postings.
- Bonuses are created in HYDRA depending on the order and operation and on the premium group. All bonuses that are stored for the premium group in HYDRA will be accounted for in the calculation. Depending on the bonus configuration, bonuses on top of the fraction stroke will be applied as "bonus" on the target time and those below as "credit note" to the actual time.
- The processing times of everyone working at the workplaces of the premium group operations (only production orders and no overhead cost orders) in the calculation period will be comprised in the denominator. In general the pro-rated personnel durations will be compensated if one person processed several operations in parallel. The sum results from the personnel postings (B-records).

Calculation of the utilization bonus

The ratio of standard time to the total runtime of all processed orders/ operations as well as of the authorized overhead cost times defines the utilization degree (NG).

$$NG = \frac{\sum_{ops} t_r + m * t_e + \sum Bonuses_{Targettime}}{\sum_{ProductionOPs} Machine\ usage + \sum_{unauth.OC-OPs} unprod.\ times - \sum Bonuses_{Actual\ time}}$$

- The utilization degree is determined per premium group.
- t_r is the set-up standard time for the operation that was stored to the HYDRA order backlog at the time of processing. The value will be used from the order postings.

- t_e is the individual standard time for the operation that was stored to the HYDRA order backlog at the time of processing. The value will be used from the order postings.
- m is the quantity that was posted during the processing of the operation in HYDRA (if necessary through several persons). Depending on the setting in the *LLE setup* this quantity can be included in scrap or not. The value will be used from the order postings.
- The machine occupation times will be added up via all production orders. The sum results from the order postings.
- Unproductive times result from postings to a (permanent) overhead cost order of general validity (e.g. cleaning of machines, lack of orders), or from machine waiting period postings. All these times will **only then** be taken into account in balancing when they are posted to the resource performance account **U8**. For the machine waiting period there is no overhead cost (OC) order in HYDRA.
 - In the default HYDRA settings the **machine waiting period** can be configured thus that it will automatically be posted to the account U8 and thus be accounted for in the formula **without further action as unproductive time**.
 - For **overhead cost (OC) postings** the times can initially be posted to other RPAs so that they will **initially apply as authorized downtime** and will not show any effect in the above formula. The foreman may then **repost these times manually** to the RPA U8 to classify them as **unauthorized downtime**. Without such a reposting through the foreman the downtimes from overhead cost orders will always be deemed authorized.
- Bonuses are created in HYDRA depending on the order and operation and on the premium group. All bonuses that are stored for the premium group in HYDRA will be accounted for in the calculation. Depending on the bonus configuration, bonuses on top of the fraction stroke will be applied as "bonus" on the target time and those below as "credit note" to the actual time.
- For different downtime reasons, the foreman can issue different (permanent) overhead cost time tickets to the operator.
- For all machines belonging to a utilization premium group, only one order can be logged in to a machine. Several operations cannot be processed in parallel at these machines.

Postings table as overview

The following table shows which postings are integrated in the calculation formulas:

Posting	Utilization bonus	Incentive bonus
Production orders (order)	1) Calculation of the standard time from produced quantity of pieces and time rules ($m * t_e + t_r$) in the numerator 2) Determination of the machine usage time (denominator)	Calculation of the standard time from produced quantity of pieces and time rules ($m * t_e + t_r$) in the numerator
Production orders (person)	- Will not be processed in the incentive wage determination-	Processing times (denominator)
Overhead cost order (order)	<i>Times only to RPA U8 (e.g. by manual reposting):</i> Integrated as unproductive times in the machine usage time (denominator). <i>Times outside the RPA U8 (e.g. default usage):</i> Will not be integrated into the formula since they are perceived as authorized downtimes. These times will, however, be shown separately and also be transferred in the interface.	- Will not be processed in the incentive wage determination-
Overhead cost order (person)	- Will not be processed in the incentive wage determination-	The sum of the overhead cost durations will not be integrated into the calculation formula, but be shown in HYDRA and be transferred in the interface. Operation numbers for which the first digit is smaller than 5 will be interpreted as "premium average". Operation numbers for which the first digit starts with 5 will be interpreted as "waiting times from GK".
Waiting period posting (machine/ OP)	Will be processed as "overhead cost order (order)". The HYDRA basic settings can be used to ensure that these postings will by default be used on the RPA U8 and correspondingly be directly counted as unproductive times.	- Will not be processed in the incentive wage determination-
Waiting period posting (person)	- Will not be processed in the incentive wage determination-	- Will not be processed in the incentive wage determination-
Bonuses	<i>Depends on the configuration of the bonus reasons:</i> As <i>bonus</i> on the standard time in the numerator and/or as <i>credit note</i> to reduce the machine usage time in the denominator	<i>Depends on the configuration of the bonus reasons:</i> As <i>bonus</i> on the standard time in the numerator and/or as <i>credit note</i> to reduce the processing time in the denominator

Additional premium formulas

When the formula-based premium/ incentive wage is used, it is also possible to realize other customer-specific calculation methods and premium formula.

Both premium forms "incentive bonus" and "utilization bonus" will be executed as default calculation by HYDRA if the premium group is configured correspondingly and can be overwritten by customer-specific formulas.

Premium areas

Premium groups can be comprised to premium areas. In doing so a premium group can be assigned to several areas so that quasi hierarchical structures can be mapped.

Configuration example:

Area 001 = Group A, B, C and D

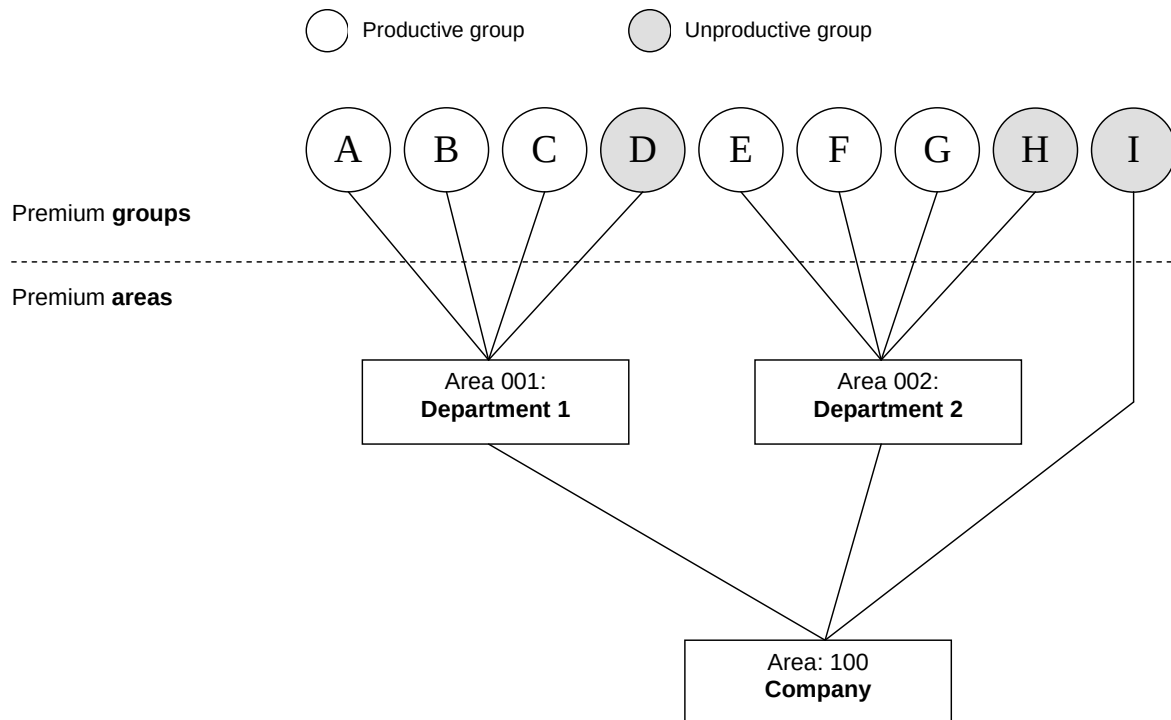
Department 1

Area 002 = Group E, F, G and H

Department 2

Area 100 = Group A, B, C, D, E, F, G, H and I company

This leads to a hierarchical structure:



Example:

The production is subdivided into several assembly lines that are paid as autonomous premium groups in group incentives. Conveyors transport the required materials to all production workplaces. Using the additional function "premium areas" it is possible to assign the conveyors to an "unproductive" group. This "unproductive" group and the assembly lines will be comprised to a premium area "production", in which the results of the individual groups will be collected and an average result be calculated. The "unproductive" conveyors will then be paid in accordance with the result from the premium area.

"Premium areas" are created as special types of premium groups in the master data. The master data of the premium group define also when a premium group must be paid according to the result of a premium area or of a different premium group.

Calculation of the area result

The result of a premium area is calculated as the result of a premium group through the wage calculation of a day. It is calculated from the day results of the assigned premium groups.

By default the individual factors such as *standard time*, *bonuses*, *unprod. time*, *duration*, *downtime*, *OC-time*, ... will be added up from the individual premium groups irrespective of the *type* of the premium group. The performance level of the premium area is calculated according to the formula

$$\text{Performance level} = \frac{\text{Standard time} + \text{bonuses}}{\text{Duration} - \text{credits}} * 100\%$$

If there is an additional function "formula-based bonus/ incentive wage" the calculation of the area result may vary depending on the customer's requests.

4 Incentive Wage Calculation - Process

Basics

The HYDRA Incentive Wage Calculation uses the data that has been recorded by a BDE or PZW system. For the different systems of premium or incentive wage, the application uses different subsets of the shop floor data recorded to calculate the incentive wages.

- Postings of operations and persons recorded by the Shop Floor Data Collection BDE in combination with the posting of times to the different resource performance accounts (RPA) performed by the Machine Data Collection MDE.
- Approved bonuses and deductions recorded by the Premium and Incentive Wage LLE
- Changes of group recorded by the Premium and Incentive Wage LLE
- Attendance, absence and break times of the employees recorded by the Personnel Time Management PZW

In addition to the transaction data above, the master data stored in the system for orders and persons and the LLE payment rules affect the data processing for the calculation of wages. For example, a configuration in the master data releases specific persons and orders for the processing in the incentive wage and bonus calculation.

Another important component of the incentive wage calculation is the complete (time) data collection in the relevant HYDRA program module. The objective is to track the complete time between clock-in and clock-out of an employee and to post the times to the relevant operations in the HYDRA system. To fulfill this requirement, HYDRA provides the waiting period processing where all times that an employee is not logged on to an operation are automatically posted to a waiting period operation.

Multiple systems exist to calculate incentive wages. On one side there are systems to calculate person-related piecework, on the other there are purely group-related bonus systems. The premium system used specifies which data recorded in HYDRA is used to calculate the incentive wage.

Using data of the Shop Floor Data Collection (BDE)

The BDE personnel postings are often used to calculate individual piecework or person-related bonuses.

To calculate group bonuses, the BDE order postings and sometimes the BDE personnel postings specify the key figures used. To distribute group results to the separate persons, the system generally uses the BDE personnel postings.

In the BDE, the following data is automatically transferred from the operation to the BDE order and personnel posting when an OP or a person is logged on (condition: the HYDRA products to calculate premium and incentive wage and/or to calculate group bonuses are licensed):

- **Single piece specification t_e**
Specified in format hours with 3 decimal places per 1000 pieces
- **Setup specification t_r**
Specified in format hours with 3 decimal places
- **Single piece specification for the production resource t_{eb}**
Specified in format hours with 3 decimal places per 1000 pieces
- **Setup specification for the production resource t_{rb}**
Specified in format hours with 3 decimal places
- **Wage type**
The wage type often specifies how a data record is used for the incentive wage calculation. Different wage types exist for piecework, bonus wage, time wage, overhead costs or other.
- **Premium group**
If the HYDRA product to calculate group bonuses is licensed, then the premium group assigned to the machine is transferred to the order and personnel postings. Optionally and as part of a customization, the premium group can be entered on the BDE terminal when the order is logged on.

In rare cases of the incentive wage calculation, the operator position (function) of the person at the machine is relevant. If required, this operator position can be entered and then processed when the person is logged on. This is a custom function. Another special feature of the incentive wage calculation can be that the person can enter a premium indicator for the activity performed when the person logs on. This premium indicator is then used to calculate the formula-based premium wage. This functionality is implemented via customization.

Using data of the Personnel Time Management (PZW)

In seldom cases, the data of the Personnel Time Management (PZW) is used to calculate the incentive wage. You can use PZW wage type postings or the totaled work day result to calculate the piecework wage or an individual bonus.

In some cases, the PZW wage type postings are assigned to the premium groups via "Change of group". The PZW wage type postings are then used to calculate the key figures for the group bonus and/or to distribute the group results to the persons.

Using data of the Premium and Incentive Wage (LLE)

On the office client, you can optionally record bonuses and changes of group (additional license or customization) for the calculation of the Premium and Incentive Wage (LLE). It depends on the system used to calculate the incentive wage whether bonuses or changes of group are included in the calculation.

You can use bonuses to manually change the premium calculation.

You can use the changes of group to assign the person-related data records of the PZW or the BDE to premium groups.

Data processing

The postings of the employees for the orders they have processed are logged in the HYDRA system. On a daily basis, these postings are transferred to the data of the incentive wage calculation to calculate the wages. Because of the above mechanism, the incentive wage calculation only includes data up to the previous day because the attendance times of the current day are only available on the next day.

If you want to correct wage-relevant data, you directly use the editing functions of the shop floor data collection and the personnel time management and you therefore directly change the output data. The incentive wage calculation then processes the data with the next run of the wage calculation (usually on the next day). The data recorded in the HYDRA system, e.g. BDE personnel postings, is only stored in the system for a specified time (35 days by default). You can only make changes to this data during this time.

For the HYDRA Premium and Incentive Wage, you can configure a longer period for the retention of wage-relevant data on the HYDRA server. If you need a longer period of data retention, these settings can be changed in the system. If you are not sure about the configuration of the data management, contact MPDV.

Overview

The wage calculation is the core function of the Premium and Incentive Wage. The system uses the BDE and PZW postings of the employees to calculate the results for the time tickets or the work day results of the persons and premium groups.

There are two options to start the wage calculation:

1. Use the HYDRA scheduler to start a daily wage calculation that calculates all work day results for persons and premium groups up to the previous day.
2. Manually start the calculation via GUI for specified days and persons or premium groups. For more information, refer to the documentation of this application on the client.

You can repeat the wage calculations as often as you like. Make sure that the data of the relevant day is still available in the system.

For further information, refer to the following documents:

- Information on the [settlement day](#).
- Information on the [performance efficiency calculation](#).
- Information on the [group bonus and premium areas](#).

Automatic recalculation of historical data

In mode *Required days until*, the wage calculation automatically recalculates all day results for persons and premium groups up to the date transferred. To this end, the system memorizes whether changes have been made to data of persons or premium groups. The relevant day results for persons and premium groups must then be recalculated. Also the automatic wage calculation entered in the scheduler starts in this mode and recalculates all relevant data up to the day before today.

The logs of the automatic wage calculations entered in the scheduler are recorded in the HYDRA system logs as application HYL_COMPUT. The manual wage calculation also creates entries here. The column *Number of data records* shows the sum total of the calculated day results for persons and premium groups. It provides an approximate evaluation of the volume that must be calculated.

The following activities trigger an automatic recalculation for the relevant days:

- Editing of bonuses (also approval/rejection) for the relevant person or premium group on the specified day
- Editing of PZW wage type postings (also approval/rejection) for the relevant person on the specified day (also if the data has been passed from external systems via interfaces)
- Recalculation of day results by the PZW labor time calculation for the relevant persons on the specified day
- Editing of BDE postings using the editing function for order-related postings of the relevant person or premium group on the specified day

Some editing activities do **not** trigger an automatic recalculation because very long periods of time or very large data volumes might be affected. In these cases, the user must manually start a wage calculation in mode *Period* for the relevant persons or premium groups, if required. Examples of such activities:

- Editing of changes of groups (these can cover a very long period)
- Editing of master data of premium groups (a period cannot be limited effectively).
- Editing of assignments of premium groups to premium areas (a period cannot be limited effectively).

Combining consecutive time tickets

This function is used to combine BDE personnel postings that have been interrupted by an automatic shift change. This is required in case of an overtime after end of shift. With a manual recording of quantities, the problem is then that the first part of the posting has a duration, but no quantity. The second part of the posting then includes the complete quantity recorded with staff logoff, but includes only a short duration. If the wage calculation calculates each of these postings separately, then the resulting performance efficiency rates would be very low with posting 1 and very high with posting 2.

To avoid this problem, the wage calculation can combine similar BDE personnel postings that are directly one after the other in time and consider them as one posting and then create one time ticket.

To activate the function to combine consecutive time tickets, go to the *Basic settings for incentive wage*.

The following conditions must be fulfilled to combine two BDE personnel postings:

Column	Condition
Machine	must be the same
Person	must be the same
Order/operation	must be the same
Cost center	must be the same
Quantity units	must be the same
Times of postings	Logoff time of the first posting must be identical to the logon time of the second posting.
Wage specifications t_e , t_r , t_{eb} , t_{rb}	must be the same
Wage type	must be the same
Premium group •	must be the same
Operator position/function	must be the same
Premium indicator	must be the same
Performance units	must be the same
User fields	Numeric user fields must be 0 or empty with both postings. Alphanumeric user fields must be identical in the first and second data record.

When two BDE postings are combined, the data is either totaled or the data of the second posting is used:

Column	Action
Quantities	are totaled.
Quantity reasons	of the second posting are used. (The second posting includes the

	information on the event that triggers the logoff).
Times of postings	are identified using the logon time of the first posting and the logoff time of the second posting.
Shift information	of the second posting is used.
Durations and resource performance accounts	are totaled.
Interruption reason	of the second posting is used. (The second posting includes the information on the event that triggers the logoff).
Performances	are totaled.

5 Wage Calculation

Overview

Menu	Human resources management → Incentive wage → Lohnberechnung
Transaction code	iwcalc
Function authorization	iwcalc.*

You use this application to start the wage calculation of the the incentive wage. The wage calculation uses postings of the employees (e.g. of the BDE and PZW) to calculate the time tickets and the work day results of persons and premium groups.

The [Incentive Wage Calculation](#) is described in a separate document.

The section below describes the manual start of the wage calculation.

In the course of the wage calculation, interrelationships can occur that are not predictable because persons can work for different premium groups at the same time. Therefore, you can hardly exclude or lock specific persons or premium groups when you start the wage calculation. Instead, the system does not allow to run different wage calculations at the same time. Only one user can start a wage calculation at a time. For the other user who starts a wage calculation, a relevant message is displayed in the list *Wage calculation*. This user must put off the start to a later point in time. This lock also affects the generation of the LLE interface file because it is not guaranteed that the data is complete when a wage calculation is run. In this case, the system does not output the usual wage calculation, but shows a warning message.

Wage calculation

Main page Profile

Request data Cancel Print preview Print all Save Help on operation Help on application Help on detail application Settings Help

Selection

Staff

☒ Evaluate staff

Person from ... to ... Name ...

Company ... Area ... Cost center ...

Premium groups

☒ Evaluate premium groups

Bonus group or area from ... to ...

Recalculate

Recalculate ☒ Required days until ☐ Fixed period

Date to 2/28/2018

Wage calculation

Drag a column here to group by that column

Result	1 2 3 ...	Text	Date
> Number of staff	9		
Number of daily personal results	19		
Number of time tickets	61		
Number of premium group days	11		
Daily personal results to be compared	4		

Selection criteria

The application provides the following selection criteria:

Evaluate staff

If this option is checked, the time tickets of the persons are recalculated. If this option is not active, only the time tickets of premium groups that have been recalculated are updated. You can limit the persons that must be recalculated using the other fields. If the selected persons worked in premium groups, it is possible that persons are recalculated who were not included in the original selection because these persons also worked in the premium groups involved.

Evaluate premium groups

If this option is enabled, the premium group results are recalculated. The user can use the other fields to further restrict the premium groups to be calculated. The system might also recalculate groups that were not included in the original selection because some of the selected persons worked in several premium groups.

Recalculate *Required days until*

All persons and premium groups selected for recalculation are recalculated up to the specified date. On the specified date, recalculation is also performed for persons and premium groups without existing wage calculation for the day. The activities that trigger an automatic recalculation of a person or premium group are described in the documentation [Incentive Wage Calculation](#).

Recalculate *Fixed period*

A recalculation of all persons and premium groups is performed for the specified period. It is recommended to restrict the data using the personnel numbers and premium groups. For reasons of security, the system limits the period that can be recalculated to a maximum of 31 days in a row.

Field descriptions**Number of staff**

Number of HR master data versions included in the original selection

Number of daily personal results

Number of recalculated personal work days

Number of time tickets

Number of recalculated personal time tickets

Number of premium group days

Number of recalculated days of premium groups

Note

Information on the completion of the wage calculation. The possible notes are described below.

Notes on the wage calculation**"Person 123456 day not computable. Reason: XYZ archiving."**

With evaluation *Required days until*: For this person, the specified day must be evaluated, but the data required of the product group specified is no longer available. The earliest possible date that can be evaluated results from the last retention date according to the data management configurations of the specified product group. If the archiving program has not yet been performed for the data, the earliest possible date is identified using the earliest data records in the product group including a safety margin of one day. The PZW data must have been calculated by the workday evaluation.

"Locked by user XX, client XX, module XX."

"Locked. Please try again later."

Another user has already started a wage calculation or an automatic wage calculation is performed in the background. Only one wage calculation at a time can be performed in the system. Try again a short time later.

"Start date limited since data not available."

With evaluation *Fixed period*: The period specified starts before the earliest possible date that can be evaluated. *Date from* has been set to the earliest possible date that can be evaluated. See also note "Day not computable since data not available."

"ERR start date after end date."

With evaluation *Fixed period*: The *Date to* is before the *Date from*. This message can also be a subsequent failure of the message "Start date limited since data not available."

"ERR Maximum nbr of 31 days exceeded"

With evaluation *Fixed period*: You can select a period covering a maximum of 31 days. This limit is used to protect the user from accidentally entering incorrect dates in the date fields.

6 Performance Level (LLE)

Synonyms

The term "Performance efficiency rate" is commonly used in the context of the piecework wage calculation.

Definition

The performance level is the measurement of performance used to calculate premium/incentive wages. It is the results from a target/actual comparison and is shown as a percentage.

Most of the time, you compare a target time with the actual time. The target time is usually specified in such a way that it doesn't exceeded by the actual time, so that the performance levels are normally above 100%. This is due to improve employee motivation. To say "I achieve more than 100%" is more motivating than saying "I have only achieved 90% of the target performance".

Performance level for piecework

The performance level is calculated in the classic piecework rate by putting the target time in relation to the actual time. Data for piecework calculation origin from the BDE personnel postings (B records). The "Wage calculation" generates from the BDE personnel postings valuated time tickets which include the performance efficiency rate.

$$performanceLevel = \frac{standardTime}{actualTime} \cdot 100\%$$

- The *actual time* in piecework depends on the basic settings for incentive wages. However, this is usually the proportionately calculated working time deduced from the logon of a person in the order data entry (B records). This is calculated proportionally for multiple machine operation. You can also use the basic settings for incentive wages to specify that the *actual time* for piecework is only made up of certain RPAs used for personnel postings.
- The target time column results from the single piece target t_e and the quantity as well as the setup time target t_r stored at the operation:
- $standardTime = \frac{quantity \cdot t_e}{1000} + t_r + bonuses$
- The t_e is usually listed in HYDRA in hours per thousand pieces [h/1000].

- The quantity is made up of the yield or yield + scrap, depending on the basic settings for incentive wages.
- The t_r is only offset if this is activated in the basic settings for *incentive wages*.
- The collected bonuses are added to each time ticket that matches the employee's personnel and order number on the date and, as a result, is totaled in the person's daily result.

The performance level is calculated for piecework time tickets and in the daily results of the persons.

A performance level is also calculated for time tickets with other time types if the required basic data is available. However, these performance levels only serve as information for the particular time tickets and are not included in further payroll accounting.

To correctly determine the target time for single piecework, the person-related collection of the produced quantities in the order data collection is a mandatory requirement. A quantity posting to the logged on persons takes place in HYDRA for the following terminal postings:

1. Log off a person from an order/workplace
2. Partial confirmations/uploads
3. Interrupting and completing orders:
 - 1) at group workplaces
 - 2) at individual workplaces: for the person who has been logged on for the longest time, but only if the option "Quantity posting to person" is activated when configuring the machine or workplace.If one of the two conditions is not fulfilled when interrupting or completing orders, the quantities are only posted to the operation, and it is not possible to determine the correct performance level for machine operators!

When using "Formula-based premium/ incentive wages", customer-specific formulas can result in different calculation methods for performance levels.

Performance level for group incentives

You can also calculate performance levels for group incentives. The [Calculation rules](#) depend on the type of group incentive.

7 Evaluation Date

Definition

The settlement date defines to which logical settlement date a data record belongs. In particular in night shifts the settlement date may be different from the posting date.

Synonyms

Settlement date, settlement day.

Example

A night shift starts on 10:00 pm and ends on 6:00 am of the next day. An order posting or a clocking record from personnel time management from 4:00 am to 6:00 am of the following day will, however, still belong to that day on which the night shift starts. The settlement date will be that day, on which the shift starts.

Assignment of the evaluation date of postings from shop floor data collection for incentive wage and labor time comparison

Both the labor time comparison and the incentive wage calculation depend on a uniform assignment of the time and labor data from time and attendance and of the recorded postings from the shop floor data collection to one consistent settlement day. In particular in case of night shifts and irregular working times this is a certain challenge since the definition of the settlement date in PZW and BDE itself is not carried out according to exactly the same rules. In general, the assignment of the working time of a person to a settlement date from PZW is leading since the subsequent day calculation carried out by PZW allows for a coherent assessment of a work day, which is not possible in the online recording of BDE. In BDE, a personnel posting is saved online without that it is known whether there will be additional personnel postings for the same work day or not.

If therefore day-wide shift models with a night shift possibility are defined in the PZW or BDE, this may lead to differences between PZE and BDE in the labor time comparison since BDE postings are assigned to a different day than the corresponding PZW time due to varying shift models and for evaluation purposes.

Please check in those cases the BDE postings also for the adjacent days in the maintenance of postings dialog. To assign the BDE personnel postings to a settlement date, two columns are used that can be displayed in the list of the maintenance of postings dialog:

Field settl. date

If the settlement date is completed, the BDE-posting will be assigned to that day. The field will be completed by the PZW labor time calculation or the editing of postings relating to orders or the wage calculation.

This means that the field for all postings of the current day will normally first be empty and only be updated on the next morning through the PZE labor time calculation and/or the wage calculation. This means that postings of the current day may initially be incorrectly assigned to the day before. This incorrect assignment will in general automatically disappear on the next day.

Please note: The PZW labor time calculation only fills out the settlement date in the personnel postings of BDE if one of the functions LLE-BPL (calculation of bonus/incentive wages), SIS-APB (comparison of labor/shop floor times) or SIS-NPB (subsequent input of labor/shop floor postings) has been licensed.

Field shift date

If the field *Settl.date* is empty, the field *shift date* will be used for assignment purposes. If the shift date does regularly not match the PZE assignment, the BDE shift models must be adapted.

Assignment by labor time calculation (personnel time management)

By default, the assignment of the BDE postings to PZE work days is made according to the following rules. Please note that these default rules may have been changed by the customizing of the incentive wage determination.

BDE personnel postings (B records)

Personnel postings are assigned to the PZE personal performance (day) if they reach into the rounded working times of the posting person in a time frame of +/- 2 hours. If a personnel posting takes several days, it will be assigned to the first possible PZW personal performance (day) since the log-off has usually been forgotten in those cases.

Order related BDE postings

Order-related BDE postings may be accounted for in the calculation of a LLE group incentive. For the BDE/PZE comparison itself they are not important.

BDE order postings (U/E/T records) are assigned to the PZE personal performance (day) of the posting person to the extent that their log-off time lies in a time frame of +/- 2 hours to the rounded working times of the posting person. If a BDE order posting takes several days, it will be assigned to the last possible PZE personal performance (day) since the posting person of the BDE order posting will be defined from the log-off event and since the person, who logs off, posts the quantities. If no personnel number is recorded in the BDE order posting, the settlement date cannot be assigned on the basis of the time and attendance function.

Assignment by wage calculation (incentive wages)

Even if Personnel Time Management (PZW) is not used in HYDRA, wage calculation as part of incentive wage determination results in BDE personnel postings being assigned to a settlement day that might deviate from the BDE shift date. This is due to the fact that a person's working day is to be considered as completed even though the person works from night shift until the next day's early shift.

Please note that these default rules may have been changed by the customizing of the incentive wage determination.

BDE personnel postings (B records)

By default, the assignment is made according to the following rules:

- 1) If a BDE personnel posting starts between 10.00 a.m. and 11.00 p.m. it pertains in any case to the date of logging in.
- 2) All BDE personnel postings starting after 11.00 p.m. and not starting later than 2.00 hours after the previous BDE personnel posting are still assigned the considered settlement day, provided they do not start later than 10.00 a.m. of the next day.
- 3) If a BDE personnel posting ends after 11.00 p.m. and has a gross duration of more than 12 hours, this posting will not be affected by this rule. This is reasonable as it avoids unintentional summarization of working days over night if persons forgot to log off.

If HYDRA Personnel Time Management (PZW) is in use, the end of day limit (11.00 p.m.) is taken dynamically from the PZW end time of the affected daily personnel performance. The end of day limit is defined based on the PZW end time. Optionally, this limit can be extended by a configurable tolerance by MPDV customizing services.

The start of day limit (10.00 a.m.), end of day limit (11.00 p.m.) and the maximum gap (2.00 hours) can be modified by MPDV customizing, if required, to be able to assign the night shift to the next day, for example (start of day limit 2.00 a.m., end of day limit 3.00 p.m.).

BDE postings relating to orders

Order-related BDE postings might be included in the calculation of an LLE group bonus. In this case, wage calculation does not assign them separately to a settlement day but takes over the assignment made in BDE (shop floor data collection) or PZW (personnel time management).

